

《计算机网络》作业

第五次作业：（英文版教材第三章 17, 18, 20, 27, 28, 29, 30, 31, 32, 33，教材第一章 15, 22，补充题 2-5）

17.(3.3.3 A Simplex Stop-and-Wait Protocol for a Noisy Channel)In the discussion of ARQ protocol in Section 3.3.3, a scenario was outlined that resulted in the receiver accepting two copies of the same frame due to a loss of acknowledgement frame. Is it possible that a receiver may accept multiple copies of the same frame when none of the frames (message or acknowledgement) are lost?

18.(3.3.3 A Simplex Stop-and-Wait Protocol for a Noisy Channel)A channel has a bit rate of 4 kbps and a propagation delay of 20 msec. For what range of frame sizes does stop-and-wait give an efficiency of at least 50%?

20. (3.4.2 A Protocol Using Go-Back-N)A 3000-km-long T1 trunk with bandwidth 1.544Mbps is used to transmit 64-byte frames using protocol 5, which uses a short ACK frame as acknowledgement. If the propagation speed is 6 μ sec/km, how many bits should the sequence numbers be?

27. (3.4.3 A Protocol Using Selective Repeat)Consider the operation of protocol 6 over a 1-Mbps perfect (i.e., error-free) line. The maximum frame size is 1000 bits. New packets are generated 1 second apart. The timeout interval is 10 msec. If the special acknowledgement timer were eliminated, unnecessary timeouts would occur. How many times would the average message be transmitted?

28. (3.4.3 A Protocol Using Selective Repeat)In protocol 6, $\text{MAX SEQ} = 2n - 1$. While this condition is obviously desirable to make efficient use of header bits, we have not demonstrated that it is essential. Does the protocol work correctly for $\text{MAX SEQ} = 4$, for example?

1-22(3.1.1 Services Provided to the Network Layer). When a file is transferred between two computers, two acknowledgement strategies are possible. In the first one, the file is chopped up into packets, which are individually acknowledged by the receiver, but the file transfer as a whole is not acknowledged. In the second one, the packets are not acknowledged individually, but the entire file is acknowledged when it arrives. Discuss these two approaches.

1-15(3.1.1 Services Provided to the Network Layer). In some networks, the data link layer handles transmission errors by requesting that damaged frames be retransmitted. If the probability of a frame's being damaged is p , what is the mean number of

transmissions required to send a frame? Assume that acknowledgements are never lost.

29. (3.4 Sliding Window Protocols) Frames of 1000 bits are sent over a 1-Mbps channel using a geostationary satellite whose propagation time from the earth is 270 msec. Acknowledgements are always piggybacked onto data frames. The headers are very short. Three-bit sequence numbers are used. What is the maximum achievable channel utilization for
- (a) Stop-and-wait?
 - (b) Protocol 5?
 - (c) Protocol 6?
30. (3.4 Sliding Window Protocols) Consider an error-free 64-kbps satellite channel used to send 512-byte data frames in one direction, with very short acknowledgements coming back the other way. What is the maximum throughput for window sizes of 1, 7, 15, and 127? The earth-satellite propagation time is 270 msec.
31. (3.4 Sliding Window Protocols) A 100-km-long cable runs at the T1 with data rate 1.544Mbps. The propagation speed in the cable is $\frac{2}{3}$ the speed of light in vacuum. How many bits fit in the cable?
32. (3.5.1 Packet over SONET) Give at least one reason why PPP uses byte stuffing instead of bit stuffing to prevent accidental flag bytes within the payload from causing confusion.
33. (3.5.1 Packet over SONET) What is the minimum overhead to send an IP packet using PPP? Count only the overhead introduced by PPP itself, not the IP header overhead. What is the maximum overhead?

补充题 2(3.4 Sliding Window Protocols). 采用 3 比特序号的选择重传 (SR) 协议, 若接收窗口为 5, 则发送窗口的最大值是多少?

补充题 3(3.4 Sliding Window Protocols). 50-kbps 的卫星信道, 往返时延为 500ms, 帧长为 1000 位, 使用捎带确认 (搭载 ACK) 的 SR 协议, 若使效率达到 50%, 序号的比特数至少是多少?

补充题 4(3.4 Sliding Window Protocols). 数据链路层采用 GBN (回退 N 步) 协议, 发送方已经发送了编号为 0-7 的帧, 当计时器超时, 若发送方只收到 0, 4, 5 号帧的确认, 则发送方需要重发的帧数是多少?

补充题 5(3.4 Sliding Window Protocols). 两台计算机的数据链路层协议实体采取滑动窗口机制利用 16kbps 的卫星信道传输长度为 128 字节的数据帧, 信道传播

时延为 270ms。若采用捎带确认（搭载 ACK）方式，且传输不会出错，

- 1) 计算使用停等协议的信道利用率；
- 2) 计算使用发送窗口为 7 的 GBN 协议的信道利用率；
- 3) 计算使用发送窗口为 15 的 GBN 协议的信道利用率；
- 4) 为使信道利用率达到最高，使用 GBN 协议时序号的比特数最少为多少位？